

Upstream and downstream

In Jenkins, the terms **upstream** and **downstream** refer to the relationship between jobs in a **pipeline** or **multi-job build** scenario. These concepts help define the flow of execution, where one job triggers or depends on another.

Upstream Jobs

An **upstream job** is a Jenkins job that **triggers** or **initiates** the execution of another job. It is the job that comes earlier in the sequence or dependency chain.

- **Example:** Job A is an upstream job for Job B if Job A triggers Job B after completing its execution.

Downstream Jobs

A **downstream job** is a Jenkins job that is **triggered** by the completion of an upstream job. It is the job that depends on the execution or outcome of another job (upstream job).

- **Example:** Job B is a downstream job of Job A if Job B is triggered automatically after Job A finishes.

Created development environmet

Configure

-  General
-  Source Code Management
-  Build Triggers
-  Build Environment
-  **Build Steps**
-  Post-build Actions

Build Steps

Execute Windows batch command ?

Command

See [the list of available environment variables](#)

```
echo "this is development env"
```

Advanced ▾

Add build step ▾

Post-build Actions

Add post-build action ▾

Save

Apply

Created test environment

Configure

-  General
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-  Build Environment
-  **Build Steps**
-  Post-build Actions

Build Steps

 **Execute Windows batch command** 

Command

See [the list of available environment variables](#)

`echo "this is test env"`

Advanced 

Add build step 

Post-build Actions

Add post-build action 

Save

Apply

Created production environment

Configure

-  General
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-  Post-build Actions

Build Steps

Execute Windows batch command

Command

See [the list of available environment variables](#)

```
echo " this is prod env"
```

Advanced 

Add build step 

Post-build Actions

Add post-build action 

Save

Apply

Configure

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-  Post-build Actions

Advanced ▾

Add build step ▾

Post-build Actions

 **Build other projects** ?

Projects to build

test

- ☒ Trigger only if build is stable
- ☐ Trigger even if the build is unstable
- ☐ Trigger even if the build fails

Add post-build action ▾

Save

Apply

Configure

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-  Post-build Actions

Advanced ▾

Add build step ▾

Post-build Actions

☰ **Build other projects** ?

Projects to build

prod,

☒ Trigger only if build is stable

☐ Trigger even if the build is unstable

☐ Trigger even if the build fails

Add post-build action ▾

Save

Apply

 Status

 Changes

 Workspace

 Build Now

 Configure

 Delete Project

 Rebuild Last

 Rename

 dev

Downstream Projects

 test

Permalinks

- [Last build \(#1\), 7 min 14 sec ago](#)
- [Last stable build \(#1\), 7 min 14 sec ago](#)
- [Last successful build \(#1\), 7 min 14 sec ago](#)
- [Last completed build \(#1\), 7 min 14 sec ago](#)

Builds

 Filter

/

Today

 #1 9:44 PM

▼



Status



Changes



Workspace



Build Now



Configure



Delete Project



Rebuild Last



Rename

Builds



Filter



Today



#1 9:44 PM



test

Upstream Projects



dev

Downstream Projects



prod

Permalinks

- [Last build \(#1\), 7 min 58 sec ago](#)
- [Last stable build \(#1\), 7 min 58 sec ago](#)
- [Last successful build \(#1\), 7 min 58 sec ago](#)
- [Last completed build \(#1\), 7 min 58 sec ago](#)



Status



Changes



Workspace



Build Now



Configure



Delete Project



Rebuild Last



Rename



prod

Upstream Projects



test

Permalinks

- [Last build \(#1\), 7 min 30 sec ago](#)
- [Last stable build \(#1\), 7 min 30 sec ago](#)
- [Last successful build \(#1\), 7 min 30 sec ago](#)
- [Last completed build \(#1\), 7 min 30 sec ago](#)

Builds



Filter



Today



#1 9:46 PM



Configure

- General
- Source Code Management
- Build Triggers**
- Build Environment
- Build Steps
- Post-build Actions

☒ Build periodically ?

Schedule ?

04 22 7 12 06

⚠ Spread load evenly by using 'H 22 7 12 06' rather than '04 22 7 12 06'

Would last have run at Saturday, 7 December, 2024, 10:04:08 pm India Standard Time; would next run at Saturday, 7 December, 2024, 10:04:08 pm India Standard Time.

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

Build Environment

☐ Delete workspace before build starts

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

≡ **Execute Windows batch command** ?

Command

See [the list of available environment variables](#)

echo "configuring cront job"

Save

Apply

 Status

 Changes

 Console Output

 Edit Build Information

 Delete build '#1'

 Timings

 Rebuild



Console Output

 Download

 Copy

[View as plain text](#)

Started by timer

Running as SYSTEM

Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\crontab

[crontab] \$ cmd /c call C:\Windows\TEMP\jenkins16970841062489447339.bat

C:\ProgramData\Jenkins\.jenkins\workspace\crontab>echo "configuring cront job"

"configuring cront job"

C:\ProgramData\Jenkins\.jenkins\workspace\crontab>exit 0

Finished: SUCCESS

Configure

- General
- Source Code Management**
- Build Triggers
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- Build Steps
- Post-build Actions

Build Triggers

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☒ Build periodically ?

Schedule ?

12 22 7 12 06

⚠ Spread load evenly by using 'H 22 7 12 06' rather than '12 22 7 12 06'

⚠ This schedule will match dates only rarely (e.g. February 29) or never (e.g. June 31), so this job may be triggered very rarely, if at all.

Advanced ▾

Add build step ▾

Configure

- General
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- Post-build Actions**

Post-build Actions

≡ **Build other projects** ?

Projects to build

test

- ☒ Trigger only if build is stable
- ☐ Trigger even if the build is unstable
- ☐ Trigger even if the build fails

After crontab set, test environment automatically triggered by dev environment

Dashboard > dev > #9 > Console Output

Status

Changes

Console Output

Edit Build Information

Delete build '#9'

Timings

Rebuild

Previous Build

✓ Console Output

Download

Copy

View as plain text

Started by timer

Running as SYSTEM

Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\dev

[dev] \$ cmd /c call C:\Windows\TEMP\jenkins6768531366902577527.bat

C:\ProgramData\Jenkins\.jenkins\workspace\dev>echo "this is development env"

"this is development env"

C:\ProgramData\Jenkins\.jenkins\workspace\dev>exit 0

Triggering a new build of test

Finished: SUCCESS